Problem

Problem · Ehrenfest's urn model. Two urns U and V contain two balls in total. At each step, one of the two balls is chosen at random (with equal probability) and moved to the other urn. Let X_n be the number of balls in U after n steps; the states are 0, 1, 2. At the start, U contains both balls.

1. Justify that (X_n) is a Markov chain, draw its probabilistic graph and write its transition matrix T (states in the order 0, 1, 2).
2. Find the unique invariant distribution π. Compare it with the distribution of the number of balls in U when each ball is placed at random in one of the two urns, independently.
3. Compute π_1, π_2, π_3, π_4. Does the sequence (π_n) converge?
4. Prove that for every $k \geq 1$, $\pi_{2k} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$ and $\pi_{2k+1} = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix}$, then that $\frac{1}{2}\left(\pi_{2k} + \pi_{2k+1}\right) = \pi$.
5. Explain, without computation, why X_n has the same parity as n.